

EXP NO : 8

DATE : 30/03/2026

PERFORMING DATA TRANSFORMATION

```
>_MONGOSH

> use salesDB
< switched to db salesDB
> db.createCollection("orders")
< { ok: 1 }
> db.orders.insertMany([
  { order_id: 1, customer: "Arun", amount: 5000, quantity: 2, status: "delivered" },
  { order_id: 2, customer: "Priya", amount: 8000, quantity: 4, status: "pending" },
  { order_id: 3, customer: "Rahul", amount: 3000, quantity: 1, status: "delivered" },
  { order_id: 4, customer: "Sneha", amount: 9000, quantity: 5, status: "shipped" },
  { order_id: 5, customer: "Kiran", amount: 7000, quantity: 3, status: "pending" }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f6144d1c4610e64798650a'),
    '1': ObjectId('69f6144d1c4610e64798650b'),
    '2': ObjectId('69f6144d1c4610e64798650c'),
    '3': ObjectId('69f6144d1c4610e64798650d'),
    '4': ObjectId('69f6144d1c4610e64798650e')
  }
}
> db.orders.aggregate([
  { $project: { customer: 1, totalPrice: { $multiply: ["$amount", "$quantity"] }, _id: 0 } }
])
< {
  customer: 'Arun',
  totalPrice: 10000
}
{
  customer: 'Priya',
  totalPrice: 32000
}
{
  customer: 'Rahul',
  totalPrice: 3000
}
{
  customer: 'Sneha',
```

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```
{
  customer: 'Sneha',
  totalPrice: 45000
}
```

```
{
  customer: 'Kiran',
  totalPrice: 21000
}
```

```
> db.orders.aggregate([
  { $project: { customer: 1, discountedAmount: { $subtract: ["$amount", 500] }, _id: 0 } }
])
```

```
< {
  customer: 'Arun',
  discountedAmount: 4500
}
```

```
{
  customer: 'Priya',
  discountedAmount: 7500
}
```

```
{
  customer: 'Rahul',
  discountedAmount: 2500
}
```

```
{
  customer: 'Sneha',
  discountedAmount: 8500
}
```

```
{
  customer: 'Kiran',
  discountedAmount: 6500
}
```

```
> db.orders.aggregate([
  { $addFields: { avgPricePerItem: { $divide: ["$amount", "$quantity"] } } }
])
```

```
< {
  _id: ObjectId('69f6144d1c4610e64798650a'),
  order_id: 1,
  customer: 'Arun',
```

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```
< {
  _id: ObjectId('69f6144d1c4610e64798650a'),
  order_id: 1,
  customer: 'Arun',
  amount: 5000,
  quantity: 2,
  status: 'delivered',
  avgPricePerItem: 2500
}
{
  _id: ObjectId('69f6144d1c4610e64798650b'),
  order_id: 2,
  customer: 'Priya',
  amount: 8000,
  quantity: 4,
  status: 'pending',
  avgPricePerItem: 2000
}
{
  _id: ObjectId('69f6144d1c4610e64798650c'),
  order_id: 3,
  customer: 'Rahul',
  amount: 3000,
  quantity: 1,
  status: 'delivered',
  avgPricePerItem: 3000
}
{
  _id: ObjectId('69f6144d1c4610e64798650d'),
  order_id: 4,
  customer: 'Sneha',
  amount: 9000,
  quantity: 5,
  status: 'shipped',
  avgPricePerItem: 1800
}
```

```

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    status: 'pending',
    avgPricePerItem: 2333.333333333335
  }
> db.orders.aggregate([
  { $project: { customer: 1, statusUpper: { $toUpper: "$status" }, _id: 0 } }
])
< {
  customer: 'Arun',
  statusUpper: 'DELIVERED'
}
{
  customer: 'Priya',
  statusUpper: 'PENDING'
}
{
  customer: 'Rahul',
  statusUpper: 'DELIVERED'
}
{
  customer: 'Sneha',
  statusUpper: 'SHIPPED'
}
{
  customer: 'Kiran',
  statusUpper: 'PENDING'
}
> db.orders.aggregate([
  { $match: { amount: { $gt: 5000 } } },
  { $project: { customer: 1, amount: 1, tax: { $multiply: ["$amount", 0.18] }, _id: 0 } },
  { $sort: { amount: -1 } }
])
< {
  customer: 'Sneha',
  amount: 9000,
  tax: 1620
}
{
  customer: 'Priya',
  amount: 8000,
  tax: 1440
}
{
  customer: 'Kiran',
  amount: 7000,
  tax: 1260
}
salesDB>

```